

Vulnerability Assessment Report

1st January 20XX

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system. NIST describes risk assessment as identifying, prioritizing, and estimating information security risks based on threats, vulnerabilities, likelihood, and impact.

Purpose

The database server is valuable because it stores customer, sales, and prospect data that supports marketing and business growth. Securing this information is important because exposure of sensitive records could damage customer trust, create legal and compliance problems, and give competitors access to business intelligence. If the server were disabled, employees working remotely would be unable to retrieve customer data, which would disrupt sales operations, delay decision-making, and affect day-to-day business performance. NIST explains that risk assessments help senior leaders determine appropriate courses of action in response to identified risks.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>cybercriminal</i>	Scan the publicly exposed database and exfiltrate sensitive customer or lead data	3	3	9

<i>Competitor</i>	<i>Attempt unauthorized access to gain business intelligence or modify records</i>	2	3	6
<i>Malicious attacker / botnet</i>	<i>Launch denial-of-service or repeated connection attempts that disrupt database availability</i>	3	3	9

Approach

This assessment focused on the fact that the database has been public-facing since launch, which makes unauthorized discovery and access more likely than in a restricted environment. The three threat sources were selected because they represent the most realistic business risks to an open database: data theft, competitive intelligence gathering, and service disruption. Their likelihood and severity were scored qualitatively based on how easily the threat could occur and how badly it could affect confidentiality, integrity, and availability. This aligns with NIST's guidance to evaluate risk through threat, vulnerability, likelihood, and impact.

Remediation Strategy

The database should no longer be open to the public internet. Access should be restricted through network controls such as IP allow-listing, VPN access, or private network segmentation so only approved employees and systems can connect. Strong authentication, role-based authorization, MFA, and detailed audit logging should be enforced to apply the principle of least privilege and support the AAA framework. Data in transit should use modern TLS, and the system should be continuously patched, monitored, and backed up so the business can reduce both breach risk and downtime. NIST's risk-management guidance emphasizes selecting and implementing controls based on risk assessment results.